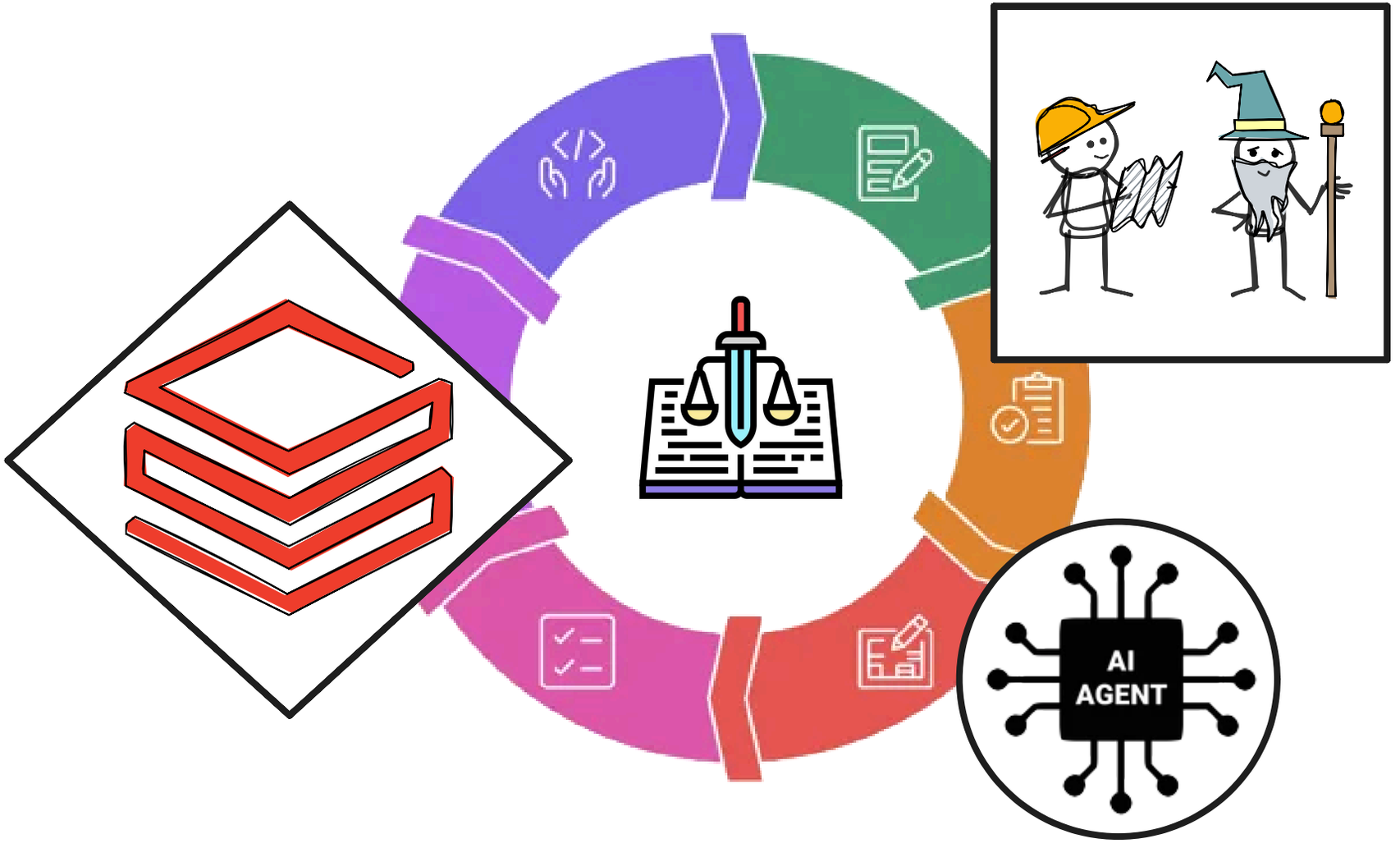
Developing on Databricks using SDD (Spec-Driven Development)



Lead Data Engineer & Data Architect
Databricks Professional Data Engineer



https://www.linkedin.com/in/arthurluz/

Lead Software Engineer
Forward-Deployed Engineer

Heitor Luz



DATA'S LIGHT

Infolog
Gestão de TI - Desenvolvimento Web

American Airlines

https://www.linkedin.com/in/heitorluz/

Warning!

This session is not about SDP!

You can ask me questions about it later 😂



databricks

Apache Spark™ Declarative Pipelines

Reliable data pipelines made easy



SDD - Spec-Driven Development Intro



Spec-driven development (SDD) is a development paradigm that uses well-crafted software requirement specifications as prompts, aided by AI coding agents, to generate executable code.



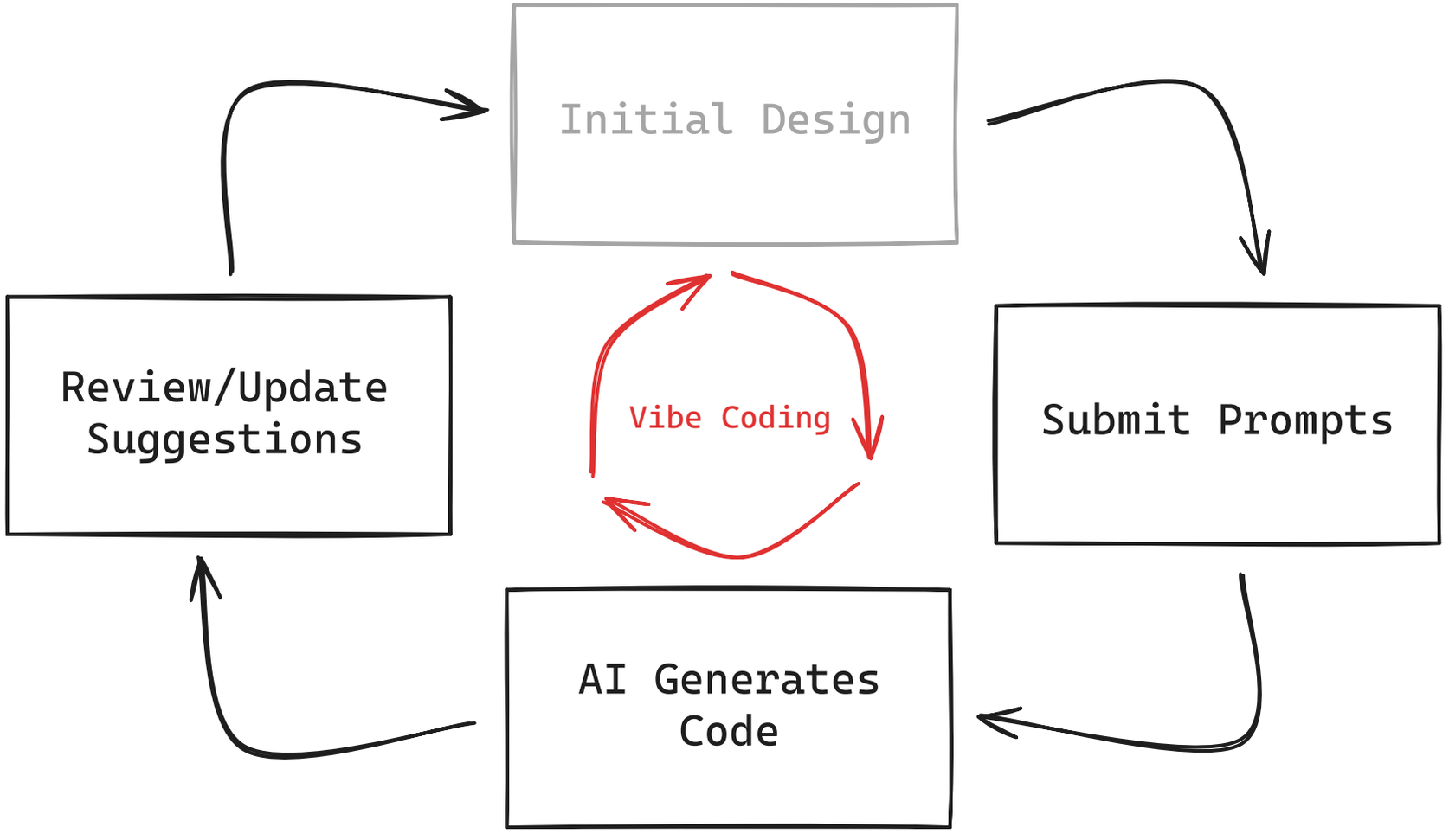
The core of SDD goes beyond vibe coding, separating the design and implementation phases. In the planning phase, requirements are first analyzed using an AI coding agent, which generates design and implementation plans.



Typically, these requirements specifications are formalized into different Markdown (.md) files. Reviewing and validating these specifications is usually an iterative process that requires a human in the loop.

https://thoughtworks.medium.com/spec-driven-development-d85995a81387

Why not using Vibe Coding?



Fast but it's fragile!

Core Principals of Spec-Driven Development

1

The spec is king

All development pivots around the spec.

2

Precision is critical

Specs are detailed enough to generate executable code.

3

AI collaboration

The AI automates, but humans maintain control.

4

The Spec is alive

If code needs to be updated, the spec is updated first.

Why it's recommended to use SDD Today?

1

AI Capabilities

AI capabilities have passed a critical threshold.

2

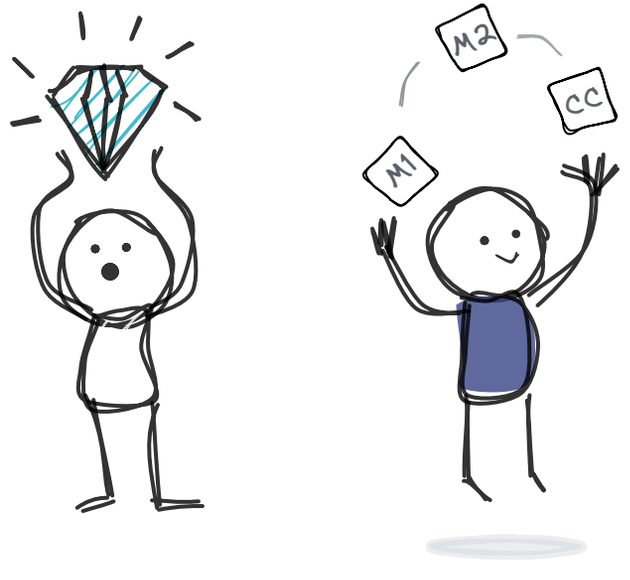
Code Complexity

Code Complexity continues to grow exponentially.

3

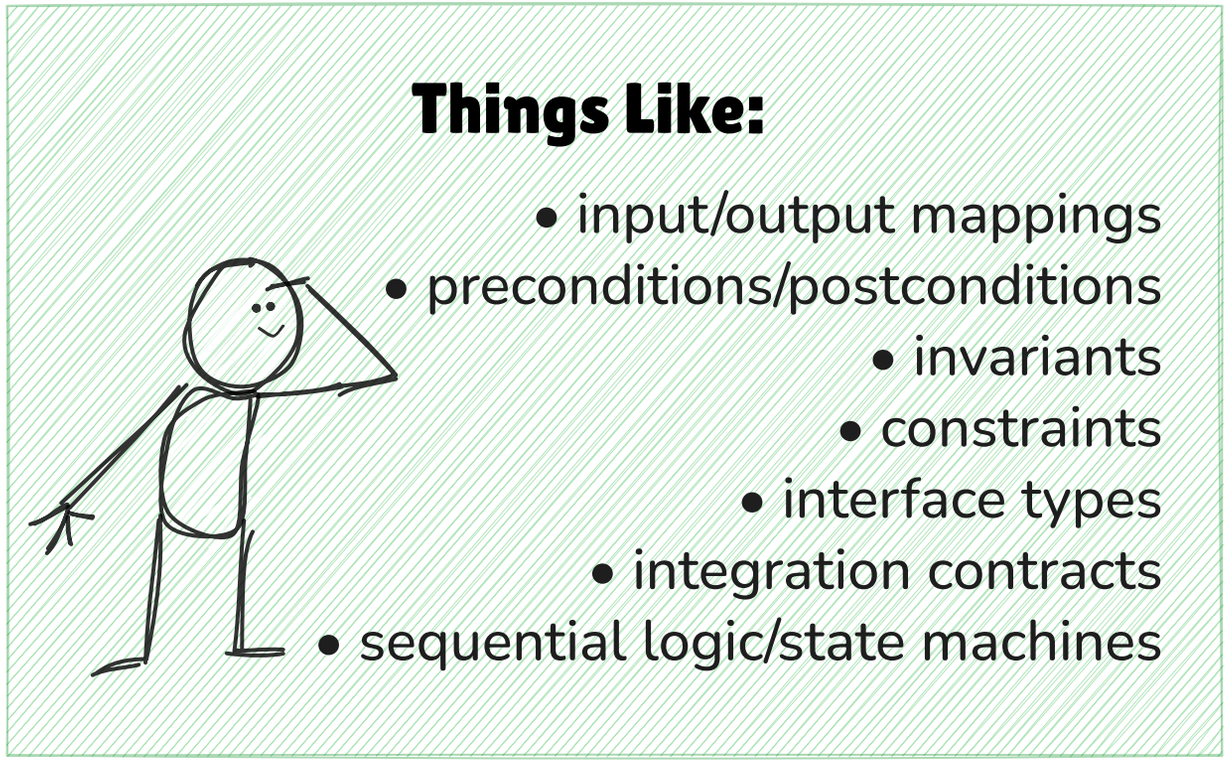
Pace of Change

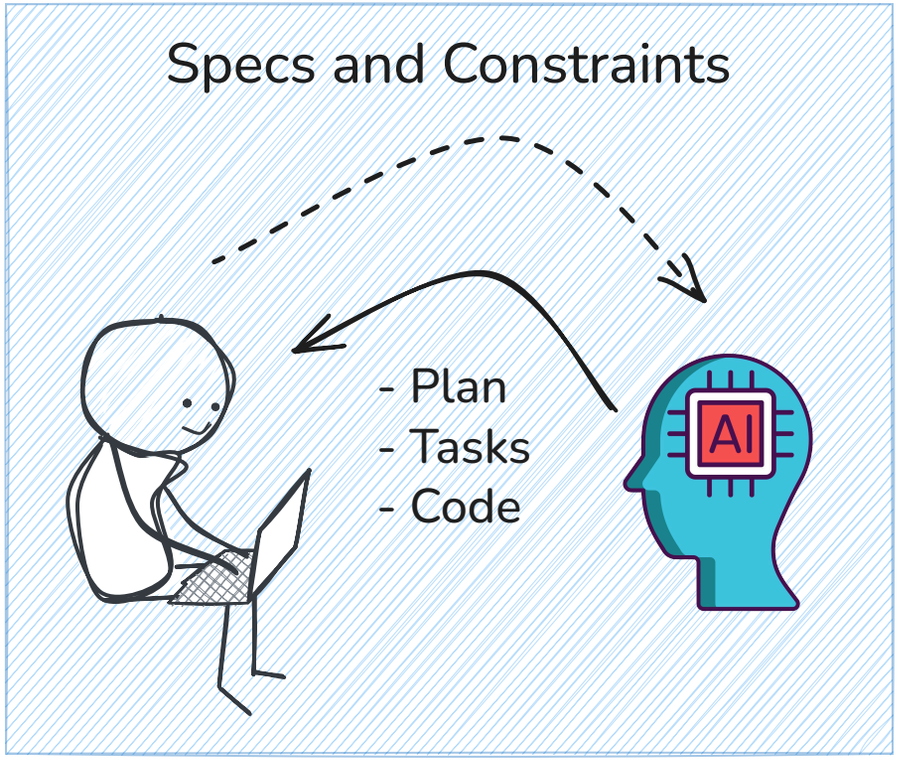
Customers demand features that represent the latest technology.



What is a Spec?

Technically, a specification should explicitly define the external behavior of the target software. Clear specifications can help reduce model hallucinations and produce more robust code.



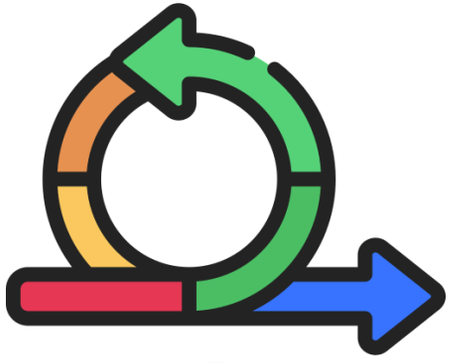


https://thoughtworks.medium.com/spec-driven-development-d85995a81387

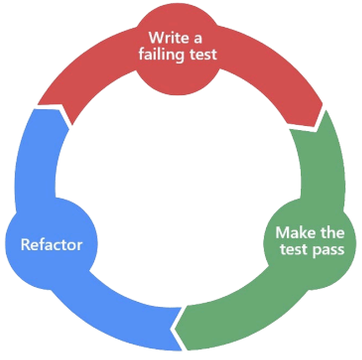
SDD as a Development Framework



Waterfall



Agile



TDD

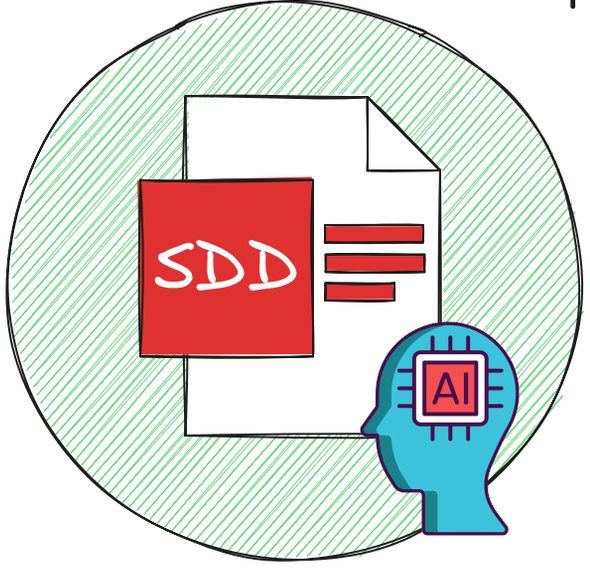


BDD

They complement each other

They complement each other

Spec is alive documentation



You can embed BDD within the Spec

GitHub Spec kit Workflow

1. /speckit:specify

PM defines the "what" and "why" for the spec. Natural Language. Focusing on user needs and functional requirements.

2. /speckit:clarify

PM + Architects resolves ambiguities by seeking clarifications and keeps the spec, plan, and tasks clear and aligned.

3. /speckit:plan

Architect defines the how - selecting frameworks, libraries, databases, and infra-estructure - and delivers a technical plan with research and quick-start guidance.

4. /speckit:tasks

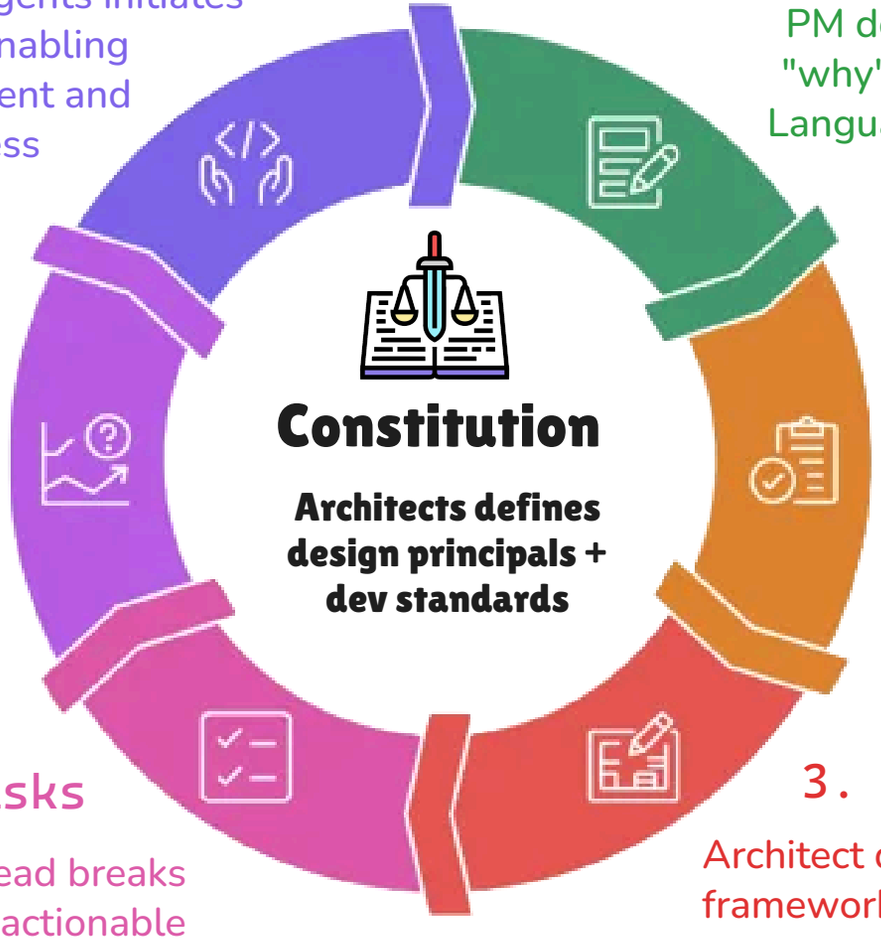
Software Eng. + Delivery Lead breaks the technical plan into clear, actionable tasks for reliable and traceable execution.

5. /speckit:analyze

Architect + PM + SWEs analyses the current spec or plan to find gaps check standards and suggest improvements.

6. /speckit:implement

SWEs + Coding Agents initiates implementation, enabling parallel development and predictable progress based on defined tasks and plan.



Understanding the Markdown artifacts

Project Tree

```
.speficy/
  |_____ memory/
     |_____ constitution.md

specs/
  |_____ 001-medallion-schemas/
     |_____ spec.md
     |_____ checklists/
        |_____ requirements.md
     |_____ plan.md
     |_____ research.md
     |_____ data-model.md
     |_____ contracts/
     |_____ quickstart.md
     |_____ tasks.md
```

1. Governance

constitution.md

Engineering Principals and constraints.

2. Requirements

spec.md

What and Why + acceptance criteria.

3. Quality

checklists/*.md

Requirements clarity and completeness.

4. Planning

plan.md + research.md

Architecture, technical choices and rationale.

5. Design & Validation

data-model.md + contracts/

Entities, schemas and interfaces.

quickstart.md

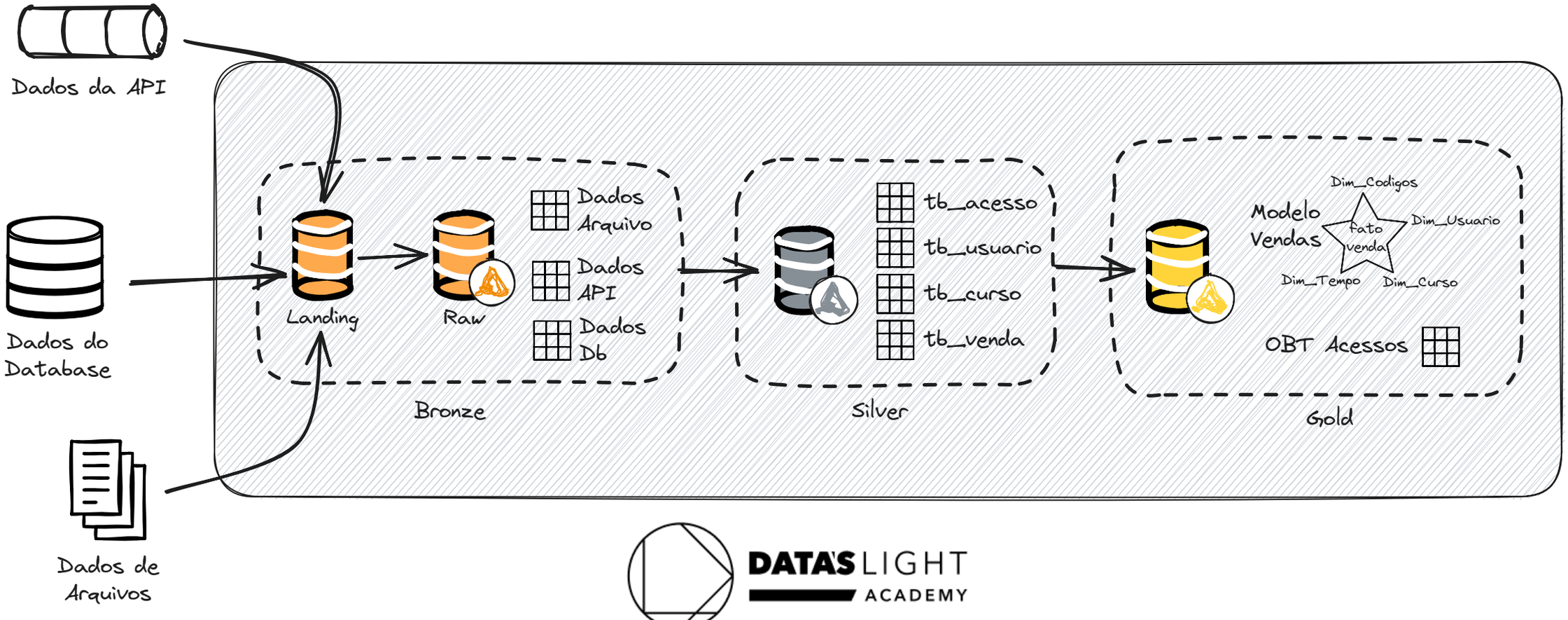
End-to-end validation scenarios.

6. Execution

tasks.md

Ordered implementation work for a person or an AI agents.

About the User-Case



DATA'S LIGHT ACADEMY

sdk do databricks	Databricks Free	Github Spec kit + Github Project	Databricks IA Tools
Claude Code	Microsoft Visual Code	Modelo Opus 5.5 High	SQL + Declarative Pipelines

Questions? Thank you!







Heitor Luz

ARTHUR LUZ

Programa Data Heroes

cupom 30% - **dbks_rj**

